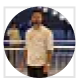
Statistics for Data Science Day 3: Measures of Central Tendency

Finding the Centre: Mean, Median, and Mode and When Each One Tells the Truth



DATA SCIENCE WITH SHIKHAR
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In Day 2, we learned how to classify data by type and scale. Now that we know what kind of data we're working with, the natural next question is: how do we summarise it? That's exactly what today's article is about.

Why Summarise Data at All?

Imagine you're handed a dataset of 10,000 customer transactions. You can't meaningfully stare at 10,000 numbers and draw conclusions. You need a single representative value that captures the "centre" of that data, a one-number summary that tells you where most of the data lives.

That's precisely what **measures of central tendency** do. The three most common are the **mean**, **median**, and **mode** and while it might seem like one of them should be enough, each exists for a good reason. By the end of this article, you'll understand exactly when to use which one and why.

> **Note:** Central tendency alone doesn't give you the full picture. To truly understand a dataset, you also need **measures of dispersion** (like variance and standard deviation) which tell you how *spread out* the data is around the centre, that's coming up in a future lesson.

The Mean (Average)

The mean, represented by the Greek letter **μ** (mu) for a population or **x̄** (x-bar) for a sample, is the most commonly used measure of central tendency. It is calculated by adding all values together and dividing by the total count.

Formula:

$$\mu = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$$

where n is the number of values and x_i represents each individual value.

Worked Example

Consider the heights (in cm) of 7 people: **[150, 155, 160, 160, 170, 175, 180]**

$$\mu = \frac{150 + 155 + 160 + 160 + 170 + 175 + 180}{7} = \frac{1150}{7} \approx 164.3 \text{ cm}$$

```
import numpy as np
data = [150, 155, 160, 160, 170, 175, 180]
print(np.mean(data)) # Output: 164.28571428571428
```

The Outlier Problem

The mean considers **every value** in the dataset which is both its strength and its weakness. Change any one value and the mean changes too. This makes it highly sensitive to **outliers** (extreme values).

Using the same dataset, let's see what happens when we add an 8th value:

8th Value Added	Mean
170 (normal)	165.0
500 (outlier)	206.25

With the outlier of 500, the mean jumps to **206.25**, a value that doesn't represent a single person in the dataset! This is the mean's major drawback, and exactly why we need the median.

> **Real-world example:** This is why average household income is often a misleading statistic. A city where 999 people earn $40,000 and one billionaire earns $1,000,000,000 will have a sky-high mean income yet the typical resident earns $40,000. The median tells the truer story.

The Median

The median is the **middle value** of a dataset when sorted in ascending order. It literally divides the dataset into two equal halves, 50% of values fall below it and 50% above it. Because it only depends on the position of values (not their magnitude), it is highly resistant to outliers.

How to Calculate

1. Sort the data in ascending order
2. If **odd** number of values → median is the middle value
3. If **even** number of values → median is the average of the two middle values

Worked Example

Using our same 7 heights, sorted: **[150, 155, 160, **160**, 170, 175, 180]**

> The middle (4th) value is **160**, that's the median.

Now add an 8th value of 170: **[150, 155, 160, 160, 170, 170, 175, 180]**

> Two middle values are 160 and 170 → Median = (160 + 170) / 2 = **165**

Now add an outlier of 500 instead: **[150, 155, 160, 160, 170, 175, 180, 500]**

> Two middle values are still 160 and 170 → Median = (160 + 170) / 2 = **165**

The median is completely unmoved by the outlier. This property is called **robustness**.

```
print(np.median(data)) # Output: 165.0
```

> **When to use median over mean:** Whenever your data contains outliers or is skewed such as income, house prices, or response times. The median gives a far more honest representation of the "typical" value.

The Mode

The mode is the **most frequently occurring value** in a dataset. Unlike the mean and median, the mode can be used on both numerical and categorical data, in fact, it is the **only** valid measure of central tendency for categorical (nominal) data.

A dataset can have:

- **No mode** - all values occur equally
- **One mode (unimodal)** - one clear most-frequent value
- **Two modes (bimodal)** - two values tied for most frequent
- **Multiple modes (multimodal)** - more than two values tied

Worked Example (Categorical)

A bank records the educational qualifications of its customers:

[Bachelor's, Master's, High School, Bachelor's, PhD, Master's, Bachelor's, Bachelor's]

Mode = Bachelor’s degree (appears 4 times - more than any other value)

This is the *only* central tendency measure that makes sense here. You cannot compute the mean or median of degree names.

Worked Example (Numerical Bimodal)

```
[12, 15, 15, 18, 20, 20, 22, 25]
```

Both **15** and **20** appear twice this is a **bimodal** distribution. Bimodal data often hints that two distinct subgroups exist in your data (e.g., two age groups, two product categories).

```python
from scipy import stats
data_cat = [15, 15, 18, 20, 20, 22]
print(stats.mode(data_cat)) # Output: ModeResult(mode=15, count=2)
#returns the smallest mode by default
```

Choosing the Right Measure

This is the most practically important part of the article. Picking the wrong measure of central tendency can lead to misleading conclusions.

Measure	Best Used When	Avoid When
Mean	Data is symmetric, no significant outliers, interval/ratio scale	Data is skewed or contains outliers
Median	Data is skewed, contains outliers, or is ordinal	You need to use the result in further calculations (e.g., variance uses mean, not median)
Mode	Data is categorical/nominal; or you want the most popular value	Data is continuous with no repeated values (mode may not exist)

A Decision Rule of Thumb

Ask yourself these questions in order:

1. **Is the data categorical?** → Use **Mode**
2. **Does the data have significant outliers or is it skewed?** → Use **Median**
3. **Neither of the above?** → Use **Mean**

The Effect of Distribution Shape

The relationship between mean, median, and mode actually reveals something important about the *shape* of your data distribution:

Distribution Shape	Relationship	What it Signals
Symmetric (Normal)	Mean ≈ Median ≈ Mode	Data is balanced around the centre
Right-skewed (Positive skew)	Mode < Median < Mean	A few very large values pulling the mean right
Left-skewed (Negative skew)	Mean < Median < Mode	A few very small values pulling the mean left

> **Practical tip:** If the mean and median of your dataset are very far apart, that's a strong signal your data is skewed and you should investigate for outliers. This is one of the first things to check during Exploratory Data Analysis (EDA).

Real-World Case Study: E-Commerce Analytics

An e-commerce platform analyses its daily order values (in USD) over a month:

- Most orders are between $20–$80
- A few corporate bulk orders of $5,000+ exist

Metric	Value	Interpretation
Mean	$312.00	Heavily inflated by bulk orders — not representative
Median	$47.00	Typical customer spends $47 — accurate picture
Mode	$29.00	The single most common order value

A junior analyst reporting the mean ($312) as the "average order value" would lead to very poor business decisions. A good data scientist immediately checks whether the mean and median are close and if they're not, the mean is hiding something.

Key Takeaway & Practical Tip

The mean, median, and mode are not competing alternatives instead they’re **complementary tools**. In any real project, compute all three and compare them. The gaps between them tell you as much about your data as the values themselves.

Quick Reference:

- Mean = sensitive to every value, best for clean symmetric data
- Median = robust to outliers, best for skewed or messy data
- Mode = the only option for categorical data, reveals the most popular value

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Up next in the series: ***Measures of Dispersion*** *— range, variance, standard deviation, and IQR. Because knowing the centre of your data is only half the story.*